

Question 2

Attempted: 0/6

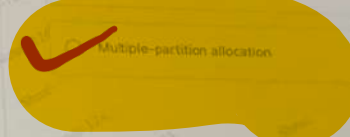
Revisit Later

Assume that you are working on a project in which you need to modify the project such that you can protect the processes from users, which means changing the operating-system code and data. Which type of allocation will you choose to accomplish the task? Select the appropriate answer from the given choices.

Select an option

Clear Response

- ☐ Module allocation
- ☐ Double-partition allocation
- ☐ Single-partition allocation



Multiple-partition allocation

Next →

Question 1

Attempted: 0/5

Revisit Later

Select an option

Clear Response

Assume that a system administrator wants to set up a file server for his/her organization. He/she wants to ensure that each department has access only to its own files and folders on the server. Which feature of the operating system can help the admin achieve this goal?

☒ Access Control Lists (ACLs)☐ User accounts☐ File ownership☐ File permissions inheritance

2. Data Structures

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2

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Attempted: 0/6

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Question 4

Revisit Later

Select an option

Clear Response

Consider that you have the given array $arr[] = \{2, 1, 2, 3, 4, 5, 6, 5, 5, 4\}$.

If you insert all the elements in an integer set, what will be the size of the set? Select the correct option from the given choices.

☐ 5.0

☐ 10.0

☐ 7.0



Question 3

Revisit Later

Select an option

Clear Response

Consider the given code:

```
1 priority_queue pq;  
2 int cnt = 0;  
3 while (pq.size() > 1) {  
4     int a = pq.top();  
5     pq.pop();  
6     int b = pq.top();  
7     pq.pop();  
8     cnt += (a + b);  
9     pq.push(a + b);  
10 }  
11 cout << cnt;
```

The code is used with two priority queues pq1, pq2 with the same array of length > 3 . For pq1, the cnt gives the value x and for pq2, it gives y. Given $x > y$. Which is the minimum and maximum heap? Analyze the given choices and select the correct option.

☒ pq1 -> max-heap, pq2 -> min-heap☐ pq1 -> min-heap, pq2 -> max-heap☐ Both are max-heap or min-heap☐ Cannot be determined

Question 5

Revisit Later

Select an option

Clear Response

Consider that you are an employee of a firm, and your boss asks you to name the layer that creates a distinct communication between Difficaultware and software. Which layer will you choose? Select the correct answer from the given choices.

☐ Microkernels

☐ Monolithic

☒ Kernel

☐ Networking

MacBook Air

Question 1

Revisit Later

Select an option

Clear Response

Consider the given dependencies:

Fish $\rightarrow$ SnakeCat $\rightarrow$ SnakeCat $\rightarrow$ Dog, Rat, Fish, Snake

What is/are the kind of Functional dependency/dependencies absent here? Select the correct answer from the given choices.

☐ Full☐ Transitive☒ Partial☐ Both A and B

3. Database Concept

Test Time: 00:50:23

Finish Test

Question 2

Attempted: 0/5

Revisit Later

Consider the given relations with functional dependencies:

PQ→R
R→S
R→T
T→U
U→P

Which aspect is present in the given relations? Select the correct answer from the given choices.

Select an option

Clear Response

☐ Multivalued attributes

☐ No atomicity

☐ Partial dependency

☒ No partial dependency

2. Data Structures

1 2 3 4 5 6

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Next Section

Question 6

Revisit Later

Attempted: 0/6

Select an option

Clear Response

Consider an array `arr[] = {18, 32, 25, 41, 63, 11, 45}`. You want to find 18 using a binary search. What will be the first element you get when you start searching and what will be the last element before you get the desired element? Select the correct option from the given choices.

☒ First element: 41, Last element: 32☐ First element: 32, Last element: 32☐ First element: 41, Last element: 18☐ First element: 25, Last element: 32

Attempted: 5/5

Question 5

Revisit Later

Select an option

Clear Response

Consider that you are the database administrator of a large online retail store. You need to ensure that the data entered into the database is consistent and accurate. Which concept of DBMS will you use to fulfill this requirement? Select the appropriate answer from the given choices.

☐ Redundancy☒ Data Integrity☐ Atomicity☐ Data Independence

Question 2

Revisit Later

Attempted: 0/0

Select an option

Clear Response

Consider that a new IP address of 26 bits has been created. You are configuring an IP address for a network device using this unique IP address. How many unique addresses can be used in this scenario? Select the correct answer from the given choices.

☒ 67108864

☐ 214748364

☐ 858993459

☐ 107374182

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3. Database Concept

1 2 3 4 5

Attempted: 0/5

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Question 4

Revisit Later

Select an option

Clear Response

As a human resources manager at a large corporation, consider that you want to store information about the company's employees in a database management system. Which resource will be most useful to ensure that each employee has a unique identifier in the database? Select the correct answer from the given choices.

☐ Composite Key☐ Candidate Key☐ Foreign Key☒ Primary Key

Attempted: 0/6

Question 5

Revisit Later

Select an option

Clear Response

Considering that you face a problem with the connection when you are trying to send an email, which layer of the OSI model may be the most likely source of the issue? Select the appropriate answer from the given choices.

☒ Transport layer☐ Network layer☐ Data Link layer☐ Physical layer

Next

Question 6

Revisit Later

Select an option

Clear Response

Consider that you are given a sent frame and a received frame as shown:

Sent - 10011010
received - 10110011

What kind of error is observed through this? Select the appropriate answer from the given choices.

☐ Burst error☒ Multiple bits error☐ Single bit error☐ Frame bit error

Question 3

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Revisit Later

Select an option

Clear Response

Assume that a data packet of size 1500 bytes is sent from a host to a destination. The IP header is 20 bytes long. The Maximum Transmission Unit (MTU) for the network is 500 bytes. How many fragments will be generated after IP fragmentation? Select the correct option from the given choices.

☐ 2.0

☐ 3.0

☒ 4.0

☐ 5.0

A OOPS

1 2 3 4 5 6 7

Attempted: 0/1

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Question 3

Revisit Later

Select an option

Clear Response

Assume that you are creating a program to simulate a library system. The program should keep track of the books in the library, including the book's title, author, ISBN number, and availability. Each book should also have the ability to be checked out or returned. Considering the given requirements, which concept in Object-Oriented Programming best represents a library book in the program? Select the appropriate answer from the given choices.

☒ Object

☐ Class

☐ Inheritance

☐ Polymorphism

Question 4

Attempted: 0/7

Revisit Later

Select an option

Clear Response

Suppose you are designing a simple banking application using Object-Oriented Programming (OOPs) concepts.

Then, identify which of the following concepts would be best suited to represent a customer's bank account?

☐ Class

☒ Object

☐ Inheritance

☐ Polymorphism



MacBook Air

Question 2

Attempted: 0/7

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Revisit Later

Select an option

Clear Response

Consider that you are a software developer building a vehicle management system for a transportation company. In order to represent different types of vehicles, you have a base class "Vehicle" with properties such as "make", "model", "year", and "price". Now, you want to create separate classes for cars, trucks, and motorcycles, each with specific properties. What is 'inheritance' in OOP? Select the correct option from the given choices.

☐ A mechanism for code duplication and complexity

☒ A mechanism for code reuse and abstraction

☐ A mechanism for code encapsulation and modularity

☐ A mechanism for code isolation and security

5. DOPS

1 2 3 4 5 6 7

Attempted: 5/7

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Question 5

Revisit Later

Clear Response

Assume that you are developing a software system for a car rental company. Which option best describes the use of abstraction in this scenario? Analyze the given choices and select the correct option.

Select an option



Abstraction is used to represent only the essential features of a car such as its make, model, and rental price while hiding unnecessary details such as its engine capacity and internal components.

☐ Abstraction is used to represent all features of a car, including its make, model, rental price, engine capacity, and internal components.

☐ Abstraction is used to hide all the features of a car, showing only essential details such as its color and rental availability.

☐ Abstraction is used to hide only essential features of a car, showing unnecessary details such as its maintenance history and previous drivers.